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ITA-0329:MOBILE COMPUTING

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CO1-AT2-Short Answer Test

Q1.)

Given,
Total spectrum=45 MHz
Bandwidth per simplex channel=25 kHz.
A full-duplex channel requires two simplex channels.
= 45,000 kHz/(2*25 kHz)
= 45,000/50=900

Q2.)
Given,
Total voice channels=560
frequency reuse cluster size=7 cells.
Channels per cell=Total channels/Cluster size
= 560/7=80

Q3.)
Given,
Number of carriers=100
Time slots per carrier=8.
Total simultaneous voice channels=Number of carriers*Time slots per carrier
= 100*8=800

Q4.)
Given,
Base-station transmit power=80 W
Handset transmit power=4 W
Power ratio(dB)=$10 \log_{10}(P_1/P_2)$
$=10 \log_{10}(80/4)$
$=10 \log_{10}(20) \approx 13.01$ dB

Q5.)
Given,
Carrier power=−60 dBmI
Interference power=−92 dBm.

C/I (dB)=Carrier power(dBm)−Interference power(dBm)
=−60−(−92)=32dB

Q6.)
Given,
Total spectrum=42 MHz
Channel bandwidth=300 kHz
Reuse cluster size=7 cells.
Total channels=42,000kHz/300kHz=140
Channels per cell=140/7=20

Q7.)
Given,
Initial received power=64 W
Path-loss exponent n=4.
$P_2 = P_1(d_1/d_2)^n$
$= 64\times(1/2)^4$
$= 64\times1/16=4W$

Q8.)
Given,
Path-loss exponent n=2
Required power scaling factor=$(R_2/R_1)^n$
$=(2R_1/R_1)^2=2^2 =4$

Q9.)
Given,
Cluster size=4 cells
Each cell is provisioned with 15 voice channels
Total channels in the cluster=4*15=60

Q10.)
Given,
Vehicle speed=100 km/h
Carrier frequency=900 MHz
Speed of light (C)≈$3*10^8$ m/s.
Convert speed(V)=100/3.6≈ 27.78 m/s
Maximum Doppler shift: fD = Vf/C
=$(27.78*900*10^6)/(3*10^8)$≈83.33 Hz

Q11.)
Given,
TDMA frame duration=5 ms

Number of equal slots=8.
Slot duration=Frame duration/Number of slots
=5ms/8=0.625 ms

Q12.)
Given,
Allocated bandwidth=24 MHZ
Carrier spacing=200 KHZ
Number of carriers=24,000KHZ/200KHZ=120

Q13.)
Given,
Transmit power=8W
Path loss=33dB
Convert transmit power to dBm:$P_t=10\ \log_{10}(8{,}000mW)\approx 39.03dBm$
Received power: $P_r=P_t-$path loss
$=39.03-33=6.03dBm$

Q14.)
Given,
Frequency=1900MHZ
Distance=2.5 km
$FSPL(dB)=32.44+20\log_{10}(f \text{ in MHz})+20\log_{10}(d \text{ in km})$
$=32.44+20\log_{10}(1900)+20\log_{10}(2.5)$
$\approx 105.97dB$

Q15.)
Given,
Number of cells=6
Each cell supports 250 simultaneous calls
Cluster capacity=Number of cells*Calls per cell
= 6*250=1,500

Q16.)
Given,
Call arrival rate=4,000 calls per second
Average call duration=3 minutes=180 seconds
Number of active calls = Arrival rate*Average duration
=4,000*180=720,000

Q17.)
Given,
Number of channels=150

Bandwidth per channel=25kHz.
Total bandwidth=150*25kHz=3,750kHz=3.75MHz

Q18.)
Given,
30 adjacent frequency bands
Each gap is 40 kHz
There are 29 gaps between 30 bands
Total guard-band bandwidth =(30−1)*40kHz
=29*40=1,160 kHz=1.16MHz

Q19.)
Given,
Vehicle speed = 90 km/h
cell radius = 1.5 km
Cell diameter = 2*1.5 = 3 km
Approximate handoffs per hour = 90 km/h/3 km = 30

Q20.)
Given,
8 regional center
40 base stations per center
250 users per base station
Total capacity = 8* 40* 250 = 80,000

Q21.)
Given,
Coverage area = 400 km²
cell radius = 400 m = 0.4 km
Approximate area per cell = R^² = 0.4^² = 0.16 km^²
Number of cells = 400 / 0.16 = 2,500

Q22.)
Given,
Total spectrum = 6 MHz
bandwidth per channel = 30 kHz
Convert 6 MHz to 6,000 kHz.
Number of channels = 6,000 / 30 = 200

Q23.)
Given,
1,200 subscribers
Each makes 4 calls per hour

Average call duration = 2.5 minutes
Traffic per subscriber = (Calls per hour*Duration in minutes) / 60
= (4*2.5) / 60 = 0.1667 Erlangs
Total traffic = 1,200* 0.1667 ≈ 200 Erlangs

Q24.)
Given,
Total bandwidth = 40 MHz
Bandwidth allocated per user = 400 kHz.
Maximum users = 40,000 / 400 = 100

Q25.)
Given,
1,200 base stations
Maximum capacity per base station = 600 simultaneous users
Total user capacity = 1,200* 600 = 720,000